Industrial Engineering and Operations Research
IIT Bombay
IE 622: Practice Problem Set 1

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1. Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space. $\{A_i\}_{i\geq 1}$ be sequence of sets such that $A_i \in \mathcal{F}\ \forall i$. Then show that $\cap_i A_i \in \mathcal{F}$.

2. Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space. A sequence of sets $\{A_n\}_{n\geq 1}$ in $\mathcal{F}$ is said to be increasing, denoted by $A_n \uparrow A$, if

$$A_1 \subseteq A_2 \subseteq \ldots\cdots \subseteq A_n \subseteq \cdots$$

   Suppose $A_n \uparrow A$ then

   (a) Show that $A \equiv \lim A_n$ is indeed an event, i.e., $\lim A_n \in \mathcal{F}$.

   (b) Show that $P(A_n) \uparrow P(A)$. That is $\lim P(A_n) = P(\lim A_n)$. Observe that we have a limiting operation of real numbers on LHS, while we have probability of limit of events on RHS. So, this property of probability measure, $P(\cdot)$, is called continuity from below property of the probability measure.

   (c) Show that (b) *implies* the countable additivity axiom of the probability measure.

3. If A is an event, show that $\sigma(A) \doteq \{A, A^c, \Omega, \phi\}$ is a σ-field.

4. Consider the probability space (Ω, P), where $\Omega = \{1, 2, 3, 4\}$ and $P(i) = 1/4$ for all $i \in \Omega$. Give an example of three events $E_1, E_2, E_3 \subset \Omega$ such that they are pairwise independent (i.e., E_i is independent of E_j whenever $i \neq j$) but not independent (i.e., $\{E_1, E_2, E_3\}$ is not independent).

5. (Memoryless property of Exp(λ)). Let X be an exponential random variable with parameter λ.

   (a) Write down the probability density function of X.

   (b) Let $t > 0$. Compute $P(X > t)$.

   (c) Let $s, t > 0$. Compute the conditional probability $P(X > t + s \mid X > s)$.

6. If X has a continuous cumulative distribution function F, then find the distribution of $F(X)$. What random variable is $F(X)$?

7. Suppose (X, Y) has a continuous density $f(x, y)$.

   (a) Find the density of $\frac{X}{Y}$. Apply to the case when (X, Y) has the standard bivariate normal distribution with density $f(x, y) = \frac{1}{2\pi} \exp\left(-\frac{x^2+y^2}{2}\right)$.

   (b) Find the distribution of $X+Y$ if (X, Y) has the standard bivariate normal distribution.

   (c) Let $U = \min\{X, Y\}$ and $V = \max\{X, Y\}$. Find the density of (U, V).

8. (a) Let X, Y be independent with $X \sim \text{Poi}(\lambda)$ and $Y \sim \text{Poi}(\mu)$. Show that $X + Y \sim \text{Poi}(\lambda + \mu)$.

   (b) Let X, Y be independent with $X \sim \text{Bin}(n, p)$ and $Y \sim \text{Bin}(m, p)$. Show that $X + Y \sim \text{Bin}(m + n, p)$.

   (c) If X, Y are integer-valued, independent and have probability mass functions (pmf) f and g, show that $X + Y$ has pmf $h(n) := \sum_{k \in \mathbb{Z}} f(k)g(n - k)$.

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IE 622: Practice Problem Set 2

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1. Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and X be a random variable. In class, we saw that the sigma algebra generated by X is defined as $\sigma(X) := \{X^{-1}(A), A \in \mathcal{B}(\mathbb{R})\}$ i.e., $\sigma(X)$ is the smallest sigma algebra that makes X measurable.

   (a) Show that $\sigma(X)$ is indeed a sigma algebra.

   (b) If $X = \mathbb{1}_B$ for some event $B \in \mathcal{F}$, what is $\sigma(X)$? What if $X = \mathbb{1}_A + \mathbb{1}_B$ where A and B are disjoint?

   (c) For two random variables X_1, X_2, how would you define $\sigma(X_1, X_2)$, the smallest sigma algebra that makes both of them measurable? What about $\sigma(X_1, X_2, \cdots, X_m)$?

2. Let $\{X_n, n = 0, 1, \ldots\}$ be a DTMC. For any $n \geq 2$ and $0 < m < n$, show that

$$\mathbb{P}(X_m = y \mid X_i = x_i \text{ for } i = 0, 1, \ldots, m-1, m+1, \ldots, n) = \mathbb{P}(X_m = y \mid X_{m-1} = x_{m-1}, X_{m+1} = x_{m+1}).$$

3. We have seen that for a stochastic process $\{X_n\}_{n \geq 0}$, for any $m \geq 0$ and $\forall\ i_1, i_2, \ldots i_m \in \mathcal{S}$ ($\mathcal{S}$ is the state space), if the probabilities

$$\mathbb{P}(X_0 = i_0, X_1 = i_1, \ldots\ldots X_m = i_m) = \lambda_0 p_{i_0 i_1} p_{i_1 i_2} \ldots p_{i_{m-1} i_m} \tag{1}$$

   then it is Markov(λ, P).

   Note that the probabilities of the form $\mathbb{P}(X_0 = i_0, X_1 = i_1, \ldots\ldots X_m = i_m)$ are some of the *fdds* (finite-dimensional distributions functions) of the Markov(λ, P). But by **Kolmogorov's consistency theorem**, all *fdds* must be defined for defining the probability law of a stochastic process. Now show that for any increasing set of finite natural numbers $n_1 \leq n_2 \leq \cdots \leq n_k$, the probabilities, $\mathbb{P}(X_{n_1} = i_{n_1}, X_{n_2} = i_{n_1}, \ldots\ldots X_{n_m} = i_{n_m})$ are also defined. **Hint:** derive it from the finite path probabilities of the form (1).

4. Let $\{X_n, n = 1, 2, \ldots\}$ be an i.i.d. integer-valued sequence. Let $X_0 = 0$. For $n \geq 1$, define:

   (i) $S_n = \sum_{m=1}^{n} X_m$

   (ii) $Y_n = X_n + X_{n-1}$,

   (iii) $Z_n = \sum_{m=1}^{n} S_m$.

   Identify which of the following are DTMCs? Justify your answer.

   (a) $\{S_n, n = 0, 1, \ldots\}$,

   (b) $\{S_{2n}, n = 0, 1, \ldots\}$,

   (c) $\{Y_n, n = 0, 1, \ldots\}$,

   (d) $\{Z_n, n = 0, 1, \ldots\}$,

(e) $\{(S_n, Z_n), n = 0, 1, \ldots\}$.

5. In a digital communication system, the receiver maintains a running average of the signal voltage to decide whether a received level corresponds to a `1` (above average) or a `0` (below average). Long runs of identical bits can cause this average to drift, leading to *baseline wander* errors.

   To mitigate this effect, a technician introduces a hardware device called `RunLengthLimiter`, which modifies a transmitted binary stream

$$(X_0, X_1, X_2, \ldots),$$

   where the bits X_i are i.i.d. Bernoulli(p).

   The device produces an output stream $(Y_0, Y_1, Y_2, \ldots)$ according to the following rules:

   (a) The first two bits are unchanged:

$$Y_0 = X_0, \qquad Y_1 = X_1.$$

   (b) For all $n \geq 2$:

   - If $Y_{n-1} = Y_{n-2}$, then the device forces

$$Y_n = 1 - Y_{n-1}.$$

   - If $Y_{n-1} \neq Y_{n-2}$, then the bit is passed through unchanged:

$$Y_n = X_n.$$

   For example, if the transmitted stream is $(X_0, \ldots, X_9) = (0\,1\,0\,0\,0\,1\,1\,0\,1\,1)$, then the output stream is $(Y_0, \ldots, Y_9) = (0\,1\,0\,0\,1\,1\,0\,0\,1\,1)$.

   Model the output process (Y_n) as a Markov chain by:

   (a) Specifying an appropriate state space.

   (b) Writing down the transition probabilities.

6. Fill in the blanks. (Use of LLMs is strongly discouraged.)

   Master Oogway: Yesterday is history, tomorrow is a mystery, but today is a gift. That is why it is called the present.
   Markov: _____
   Send in your witty responses via this form.

Industrial Engineering and Operations Research
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IE 622: Practice Problem Set 3

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and $(X_n)_{n\geq 0}$ be a Markov chain on the state space I with transition probability matrix P and initial distribution λ.

1. Show that the hitting time of a set $A \subset I$ defined as

$$H^A(\omega) = \inf\{n \geq 0 : X_n(\omega) \in A\}$$

is indeed a random variable.

2. Let $\{X_n, n = 0, 1, \ldots, \}$ be a discrete-time random process, and let S and T be stopping times for $\{X_n\}$. Which of the following are stopping times?

   (a) $\min\{S, T\}$,

   (b) $S - 1$,

   (c) $T + 1$,

   (d) U^j = time of last visit to state j,

   (e) a constant time, say k.

3. Let $\{X_n, n = 0, 1, \ldots\}$ be an irreducible DTMC on a finite state space $\mathcal{X}$. Let $A \subset \mathcal{X}$ be a nonempty subset of the state space. Let $\{T_m, m = 1, 2, \ldots\}$ denote the successive entrance times to A :

$$T_m = \min\{n > T_{m-1} : X_{n-1} \notin A, X_n \in A\},$$

with the convention $T_0 = 0$.

   (a) Show that, for each $m \geq 1$, T_m is a stopping time.

   (b) Argue that $\{X_{T_m}, m = 1, 2, \ldots\}$ is a DTMC. What is its state space?

4. Consider a DTMC $\{X_n\}$ with the following TPM. Find the communicating classes and identify which among them are closed. Is the DTMC irreducible?

   (a)

$$\begin{pmatrix} 0.1 & 0.8 & 0 & 0 & 0.1 \\ 0.3 & 0.7 & 0 & 0 & 0 \\ 0 & 0 & 0.9 & 0.1 & 0 \\ 0 & 0 & 0.4 & 0.6 & 0 \\ 0 & 0.1 & 0.8 & 0.1 & 0 \end{pmatrix}$$

(b)

$$\begin{pmatrix} 0.5 & 0.5 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0.5 & 0 & 0 & 0.25 & 0.25 & 0 \\ 0 & 0 & 0 & 0.4 & 0.6 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0) \end{pmatrix}$$

5. (a) In class, C was defined as a closed communicating class if for all $i \in C$ and $j \notin C$, $p_{ij} = 0$. Show that this definition is equivalent to the following statement: For every $i \in C$, if $i \to j$ then $j \in C$.

   (b) (JN 1.2.2) Show that every transition matrix on a finite state-space has at least one closed communicating class. Find an example of a transition matrix with no closed communicating class.

6. Fill in the blanks. (Use of LLMs is strongly discouraged.)

   Markov: Past and future are independent given the present.
   Strong Markov: Stop your rant! I can show it holds for any stopping time.
   Swami Vivekananda: Arise, awake, and stop not till the goal is reached.
   Strong Markov / Markov: _____
   Send in your responses via this form.

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IE 622: Practice Problem Set 4

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1. (Takis-#10) Smith is in jail and has 3 dollars; he can get out on bail if he has 8 dollars. A guard agrees to make a series of bets with him. If Smith bets A dollars, he wins A dollars with probability 0.4 and loses A dollars with probability 0.6. Find the probability that he wins 8 dollars before losing all of his money if
   (a) he bets 1 dollar each time (timid strategy).
   (b) he bets, each time, as much as possible but not more than necessary to bring his fortune up to 8 dollars (bold strategy).
   (c) Which strategy gives Smith the better chance of getting out of jail?

2. (Takis-#12) A fair coin is tossed repeatedly and independently. Find the expected number of tosses until the following patterns appear:
   (a) HTH.
   (b) HHH.
   (c) THTH.

3. Reading assignment: [JN 1.6], Random walk slides and Errata for Theorem 8 on Moodle.

4. (RD Thm 6.4.3) Define $f_{ij} = \mathbb{P}(T_j < \infty | X_0 = i)$ where $T_j = \inf\{n \geq 1 : X_n = j\}$. If state $i \in I$ is recurrent and $f_{ij} > 0$, then show that j is recurrent and $f_{ji} = 1$.

5. Prove that state i is transient if and only if $P(X_n = i \text{ for infinitely many } n) = 0$. Note that, in class, we defined transience in a different way.

6. Let $T_i = \min\{n \geq 1 : X_n = i\}$ and define $f_{i,j} = P_i(T_j < \infty)$. Let $N_j = \sum_{n=1}^{\infty} 1_{\{X_n = j\}}$ denote the number of visits to j. Show that
$$E_i[N_j] = \frac{f_{i,j}}{1 - f_{j,j}}.$$

7. Consider a DTMC $\{X_n\}$ with the following TPMs. Classify the vertices as transient and recurrent and find a stationary distribution.
   (a)
$$\begin{pmatrix} 0.1 & 0.8 & 0 & 0 & 0.1 \\ 0.3 & 0.7 & 0 & 0 & 0 \\ 0 & 0 & 0.9 & 0.1 & 0 \\ 0 & 0 & 0.4 & 0.6 & 0 \\ 0 & 0.1 & 0.8 & 0.1 & 0 \end{pmatrix}$$

(b)

$$\begin{pmatrix} 0.5 & 0.5 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0.5 & 0 & 0 & 0.25 & 0.25 & 0 \\ 0 & 0 & 0 & 0.4 & 0.6 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 \end{pmatrix}$$

8. (Takis-#5) I have 4 umbrellas, some at home, some in the office. I keep moving between home and office. I take an umbrella with me only if it rains. If it does not rain I leave the umbrella behind (at home or in the office). It may happen that all umbrellas are in one place, I am at the other, it starts raining and must leave, so I get wet.

   - If the probability of rain is p, what is the probability that I get wet?
   - Current estimates show that $p = 0.6$ in Mumbai. How many umbrellas should I have so that, if I follow the strategy above, the probability I get wet is less than 0.1?

9. (Takis-#8) Consider the knight's tour on a chess board: A knight selects one of the next positions at random independently of the past.

   (a) Why is this process a Markov chain?

   (b) What is the state space?

   (c) Is it irreducible?

   (d) Find the stationary distribution. Give an interpretation of it: what does it mean, physically?

   (e) Which are the most likely states in steady-state? Which are the least likely ones?

10. Determine whether or not the random walk on the integers with $p_{i,i+2} = p, p_{i,i-1} = 1 - p$ for all i is recurrent.

11. Show by example that Markov chains that are not irreducible may have many different stationary distributions.

12. Fill in the blanks. (Use of LLMs is strongly discouraged.)

    Newton: An object at rest remains stationary unless acted upon by an external force.
    Knight on the chessboard / Sir Jamie Lannister:————-

    Write a response in this form.

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1. (Elevator Markov chain) Consider the DTMC with state space $\{0, 1, \ldots, \}$ and transition probabilities

$$p_{i,j} = \begin{cases} 1 & \text{if } i \geq 1 \text{ and } j = i - 1, \\ q_k & \text{if } i = 0 \text{ and } j = k \geq 1, \\ 0 & \text{otherwise,} \end{cases}$$

where $\{q_i, i = 1, 2, \ldots\}$ are such that $0 < q_i < 1$ and $\sum_{i=1}^{\infty} q_i = 1$. Show that:

(a) This DTMC is recurrent.

(b) It is positive recurrent if and only if $\sum_{i=1}^{\infty} i q_i < \infty$.

(If Hotel California was infinitely tall, you can check out anytime you like but you can never leave unless $\sum_{i=1}^{\infty} i q_i < \infty$.)

2. Consider the following transition probability matrix:

$$P = \begin{array}{c} \\ 1 \\ 2 \\ 3 \\ 4 \\ 5 \end{array} \begin{array}{c} \begin{array}{ccccc} 1 & 2 & 3 & 4 & 5 \end{array} \\ \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 2/3 & 0 & 1/3 & 0 \\ 1/8 & 1/4 & 5/8 & 0 & 0 \\ 0 & 1/6 & 0 & 5/6 & 0 \\ 1/3 & 0 & 1/3 & 0 & 1/3 \end{pmatrix} \end{array}$$

Find $\lim_{n\to\infty} p_{ij}^{(n)}$ for all states i and j.

3. Consider the Markov chain $(X_n)_{n\geq 0}$ with state space $I = \{0, 1, 2\}$ and transition matrix

$$P = \begin{pmatrix} r & 1-r & 0 \\ q & 0 & p \\ 0 & s & 1-s \end{pmatrix}, \qquad r, p, q, s \in (0, 1), \quad p + q = 1,$$

and initial distribution $\lambda_0 = 1$. Let $T_0 := \inf\{n \geq 1 : X_n = 0\}$. For $j \in \{0, 1, 2\}$, define

$$N_j := \sum_{k=0}^{T_0 - 1} \mathbf{1}_{\{X_k = j\}},$$

the number of visits to state j before returning to 0.

(a) Compute $\gamma_0^0 = \mathbb{E}_0[N_0]$, $\gamma_1^0 = \mathbb{E}_0[N_1]$, and $\gamma_2^0 = \mathbb{E}_0[N_2]$.

(b) Compute the stationary distribution π of the chain and verify that

$$\pi_j = \frac{\gamma_j^0}{\sum_j \gamma_j^0}, \qquad j = 0, 1, 2.$$

4. (Problem 1.74 in Durrett) Consider the DTMC with state space $\{0, 1, \ldots\}$ with transition probabilities $p_{i,i+1} = q_i = 1 - p_{i,0}$ for $i \geq 0$. Find conditions on $\{q_i, i = 0, 1, \ldots\}$ under which this DTMC is (i) recurrent and (ii) positive recurrent. Find the unique stationary distribution when it is positive recurrent.

5. (SR 4.2.2) Show that if $i \leftrightarrow j$, then $d(i) = d(j)$, where $d(.)$ is the period of the state.

6. Consider a discrete-time Markov chain $\{X_n, n \geq 0\}$ on the state space $S = \{1, 2, 3, 4, 5\}$ with the following one-step transition matrices P:

(a)

$$\begin{pmatrix} 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 1/2 & 0 & 0 & 1/2 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 & 0 \end{pmatrix}$$

(b)

$$\begin{pmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0.5 & 0.5 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1/3 & 2/3 \\ 0 & 0 & 0 & 2/3 & 1/3 \end{pmatrix}$$

Find the period of all states. Are any of them aperiodic?

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1. Let $\{Y_n, n \geq 1\}$ be a sequence of i.i.d. random variables with $P(Y_n = 2) = p = 1 - P(Y_n = -1)$. The process $\{X_n, n \geq 0\}$ is defined as follows:

$$X_0 = 0$$

$$X_{n+1} = (X_n + Y_{n+1})^+$$

where $(\cdot)^+$ denotes the positive part, i.e., $x^+ = \max(0, x)$.

   a. Show that $\{X_n\}$ is a DTMC.
   b. Display its transition probability diagram.
   c. Obtain a sufficient condition for the positive recurrence of $\{X_n\}$.
   d. Let Z_n be the number of times $\{X_n\}$ visits 0 over the interval $[0, n]$. How will you obtain

$$\lim_{n\to\infty} \frac{1}{n} E(Z_n)$$

   for the parameter values obtained in part (c)? (No need to derive the actual expressions.)

2. Let $\{X_n, n \geq 0\}$ be a Discrete-Time Markov Chain (DTMC) on the state space $S = \{0, 1, 2, \dots\}$ such that:

   - $p_{0i} = \frac{\alpha}{i^2}$ for all $i \geq 1$, with α such that $\sum_{i=1}^{\infty} \frac{\alpha}{i^2} = 1$.
   - For $i \geq 1$, $p_{i,i+1} = p$ and $p_{i,i-1} = q$ (where $q = 1 - p$), with $p < 1/2$.

   a. Display the transition probability diagram.
   b. Use the Foster-Lyapunov theorem to show that $f_{00} = 1$.
   c. Show that there is an $\epsilon > 0$ such that $E(X_{n+1} - X_n \mid X_n = i) < -\epsilon$ for all $i \geq 1$, and yet the DTMC is null recurrent. [Hint: Use one-step recursion for m_0 and a lower bound for $\mathbb{E}_i[T_0] = \sum_{k=1}^{\infty} k\mathbb{P}_i(T_0 = k)$.]
   d. Does (c) violate the Lyapunov condition for positive recurrence? Justify

3. Let $(X_n)_{n\geq 0} \sim$ Markov (π, P) where π is the stationary distribution of P. Time reversibility says that for any $N \geq 1$, the reversed chain $(X_{N-n})_{n\geq 0}$ has the **same** transition probabilities as the forward chain $(X_n)_{n\geq 0}$. In each of the following cases determine whether the stochastic matrix is reversible.

   (a)

$$P = \begin{pmatrix} 0 & p & 1-p \\ 1-p & 0 & p \\ p & 1-p & 0 \end{pmatrix}.$$

(b) $I = \{0, 1, 2, \cdots, N\}$ and $p_{ij} = 0$ if $|j - i| \geq 2$.

(c) $p_{ij} = p_{ji}$ for all $i, j \in S$, where S is the state space.

(d) $I' = \{0, 1, 2, \dots\}$ and $p_{01} = 1, p_{i,i+1} = p, p_{i,i-1} = 1 - p$ for $i \geq 1$.

4. Let $G = (V, E)$ be a connected, undirected graph with vertex set V and edge set E. A simple random walk on G is a Markov chain $\{X_n\}_{t \geq 0}$ with transition probabilities:

$$P_{ij} = \mathbb{P}[X_{n+1} = j \mid X_n = i] = \begin{cases} \frac{1}{d_i} & \text{if } (i, j) \in E \\ 0 & \text{otherwise} \end{cases}$$

where d_i is the degree of vertex i. Define the stationary distribution π by $\pi_i = \frac{d_i}{2|E|}$ for all $i \in V$.

(a) Show that the random walk is time-reversible with respect to the stationary distribution π. Specifically, prove the *detailed balance condition*:

$$\pi_i P_{ij} = \pi_j P_{ji} \quad \forall i, j \in V$$

(b) Compute the stationary distribution for the random walk of the knight on a chessboard, i.e., at each square it chooses among one of its legal moves uniformly at random.

5. **Hermione's borrowed Time-turner:**

I mark the hours, every one,
Nor have I yet outrun the Sun.
My use and value, unto you,
Are gauged by what you have to do.

Hermione: Dumbledore said we must not be seen! We have to go back to the spot where we were, and stay there until we leave, or we could kill our past selves!
Time-reversed Markov:——————

Send in your answers via this Form

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1. Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space.

   (a) Show that countable union and countable intersection of almost sure events is almost sure.

   (b) Let A, B be two events where B occurs a.s. Show that $\mathbb{P}(A \mid B) = \mathbb{P}(A)$. This means that conditioning on an almost sure event does not change the probability of an event.

   (c) Show that if X is a random variable with a finite expectation i.e. $\mathbb{E}[|X|] < \infty$, then $|X| < \infty$ a.s.

2. Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space. Let $(X_n)_{n\geq 1}, X$ be random variables on it and $(A_n)_{n\geq 1}$ be a sequence of events in $\mathcal{F}$. Show that:

   (a) $\limsup A_n, \liminf A_n \in \mathcal{F}$

   (b) $\liminf A_n \subseteq \limsup A_n$

   (c) $\{\omega \in \Omega : \lim_{n\to\infty} X_n(\omega) = X(\omega)\} \in \mathcal{F}$

   (d) For any $\omega \in \Omega$, $\mathbb{1}_{\limsup A_n}(\omega) = \limsup \mathbb{1}_{A_n}(\omega)$

3. (**The Infinite Monkeys theorem**) Suppose a monkey types on a simplified keyboard with only uppercase letters. The monkey types independently and uniformly at random. Show that it will type the word 'BANANA' infinitely often.

4. (**Online dating**) (taken from IITM Krishna Jagannathan EE510 course) On a certain day, Alice decides that she will start looking for a potential life partner on an online dating portal. She decides that everyday, she will pick a guy uniformly at random from among the male members of the dating portal, and go out on a date with him. What Alice does not know, is that her neighbor Bob is interested in dating her. Being of a shy disposition, Bob decides that he will not ask Alice out himself. Instead, he decides that he will go out on a date with Alice only on the days that Alice happens to pick him from the dating portal, of which he is already a member. For the first two parts, assume that 50 new male members and 40 new female members join the dating portal everyday.

   (a) What is the probability that Alice and Bob would have a date on the nth day? Do you think Bob and Alice would eventually stop meeting? Justify your answer, clearly stating any additional assumptions.

   (b) Now suppose that Bob also picks a girl uniformly at random everyday, from among the female members of the portal, and that Alice behaves exactly as before. Assume also that Bob and Alice will meet on a given day if and only if they both happen to pick each other. In this case, do you think Bob and Alice would eventually stop meeting?

(c) For this part, suppose that Alice and Bob behave as in part (a), i.e.,Alice picks a guy uniformly at random, but Bob is only interested in dating Alice. However, the number of male members in the portal increases by 1 percent everyday. Do you think Bob and Alice would eventually stop meeting?

5. Demonstrate with an example (using random variables defined on a common probability space) that convergence in distribution does not imply convergence in probability. That is, come up with a sequence $\{Y_n\}$ and a random variable Y (all defined on some common probability space) such that $Y_n \xrightarrow{d} Y$ but $Y_n \xrightarrow{\mathbb{P}} Y$.

6. (a) Let $\{X_n\}$ be i.i.d. with finite mean. Let $Y_n = \mathbf{1}_{\{|X_n| \leq n\}} X_n$. Show that $E[Y_n] \to E[X_1]$.

   (b) Does the conclusion of part (a) hold if we remove the i.i.d. assumption? That is, if $\{X_n\}$ is a sequence of random variables with $EX_n = EX_1 < \infty$ for all n, is it true that $E[Y_n] \to E[X_1]$? Justify.

7. Shakespeare and Monkey
   Shakespeare: "The fool doth think he is wise, but the wise man knows himself to be a fool" — *As You Like It*
   Monkey: I typed that even before you said it. How do you like that?
   Shakespeare: "To be, or not to be, that is the question" — *Hamlet*
   Borel-Cantelli / Monkey:_____

   Send in your replies via this Form.

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1. Let (X_t) be a continuous-time random process taking values in a discrete set $\mathcal{S}$. Assume that the sample paths of (X_t) are right continuous. Express the probability $P(X_t = i \text{ for all } t \in [t_1, t_2])$ in terms of the finite dimensional distributions of (X_t).

2. (Properties of exponential random variables). Suppose $\{S_i\}$ are independent exponential random variables with parameter $\{\lambda_i\}$. Show the following:

   (a) $\lambda_1 S_1$ has distribution Exp(1).

   (b) $\min\{S_1, \ldots, S_n\}$ has distribution $\text{Exp}(\lambda_1 + \cdots + \lambda_n)$. That is, for any t,
   $$P(\min\{S_1, \ldots, S_n\} > t) = \exp\{-(\lambda_1 + \cdots + \lambda_n)t\}.$$

   (c)
   $$P(S_1 < \min\{S_2, \ldots, S_n\} \mid \min\{S_1, \ldots, S_n\} > t) = \frac{\lambda_1}{\lambda_1 + \cdots + \lambda_n} \quad \text{for all } t.$$
   In particular, for any t, the events $\{S_1 < \min\{S_2, \ldots, S_n\}\}$ and $\{\min\{S_1, \ldots, S_n\} > t\}$ are independent.

   (d) If $\sum_{n=1}^{\infty} \frac{1}{\lambda_n} < \infty$ then $P(\sum_{n=1}^{\infty} S_n < \infty) = 1$.

   (e) If $\sum_{n=1}^{\infty} \frac{1}{\lambda_n} = \infty$ then $P(\sum_{n=1}^{\infty} S_n = \infty) = 1$.

3. Let (X_t) be a Poisson process of rate λ. Let $p_0(t) = P(X_t = 0)$. Show that p_0 satisfies the ODE
   $$p_0'(t) = -\lambda p_0(t), \quad t \geq 0.$$

4. Let (X_t) be a counting process that has independent and stationary increments with X_t having distribution Poisson(λt) for all $t \geq 0$. Compute $P(X_{t_1} = x_1, X_{t_2} = x_2, \ldots, X_{t_n} = x_n)$ for every $0 \leq t_1 < t_2 < \cdots t_n$ and $0 \leq x_1 \leq \cdots \leq x_n$ and show that these finite-dimensional distributions are completely determined.

5. Suppose (N_t) is a Poisson process with rate 2. Compute the conditional probabilities

   (a) $P(N_3 = 4 \mid N_1 = 1)$,

   (b) $P(N_1 = 1 \mid N_3 = 4)$.

6. Let (N_t) be a Poisson process of rate λ. Show that
   $$\text{Cov}(N_t, N_s) = \lambda \min\{t, s\}.$$
   [Note: For two random variables X and Y, the covariance is defined by $\text{Cov}(X, Y) = E[(X - E[X])(Y - E[Y])]$.]

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1. (Reading assignment). Theorem 2.4.6 in Norris.

2. (Exercise given in class). Consider the setting of splitting a Poisson process (X_t) discussed in class: Let $\{\theta_n\}$ be i.i.d. Bernoulli(p) random variables, independent of (X_t), and define

$$X_t^1 = \sum_{n=1}^{\infty} \theta_n 1\{J_n \le t\}, \qquad \text{and } X_t^2 = \sum_{n=1}^{\infty} (1-\theta_n) 1\{J_n \le t\}, \quad t \ge 0.$$

   Show that the processes (X_t^1) and (X_t^2) are independent.

3. Show directly from the infinitesimal definition (i.e., the second characterization done in class) of a Poisson process of rate λ that the first jump time J_1 is exponentially distributed with parameter λ.

4. Demonstrate with an example that the superposition of two dependent Poisson processes need not be a Poisson process.

5. Let (X_t) be a Poisson process of rate λ. Show that

$$\frac{X_t}{t} \to \lambda \text{ as } t \to \infty, \text{ almost surely.}$$

6. Consider two independent Poisson processes $\{N_t^{(1)}\}$ and $\{N_t^{(2)}\}$ with rates λ_1 and λ_2, respectively. Compute the probability that the two-dimensional process $\{(N_t^{(1)}, N_t^{(2)})\}$ ever visits the point (i, j).

7. Two people are working in a small office selling shares in a mutual fund. Each is either on the phone or not. For $i = 1, 2$, suppose that salesman i is on the phone for an exponential amount of time with rate μ_i and then off the phone for an exponential amount of time with rate λ_i. Formulate a CTMC model for this system with state space $\{0, 1, 2, 12\}$ where the state indicates who is on the phone.

8. Consider a function $\lambda : \mathbb{Z}_+ \to [0, \infty)$, where $\mathbb{Z}_+ = \{0, 1, 2, \ldots\}$ is the set of nonnegative integers. Let (X_t) be a CTMC on $\mathbb{Z}_+$ whose off-diagonal entries of the rate matrix are given by $q_{i,i+1} = \lambda(i)$ for $i \ge 0$ and $q_{i,j} = 0$ if $j \ne i+1$. Define the random variable $\zeta = \lim_{n\to\infty} J_n$, where J_n is the time of the nth jump in (X_t). Note that ζ is well defined since J_n are increasing, and ζ is called the "time of explosion." Specify sufficient conditions on the function λ to have (i) $P(\zeta < \infty) = 1$, and (ii) $P(\zeta = \infty) = 1$.

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1. Suppose a queue with a buffer capacity of K is fed by Poisson arrivals with i.i.d. service times that are exponentially distributed and independent of arrivals. When an incoming customer finds that the buffer is not full (i.e., at most $K-1$ customers are present in the buffer), then it joins the queue; otherwise, it does not join the queue and leaves the system without getting service.

   (a) Model the dynamics of this queue as a stochastic process.

   (b) Under stationarity, what is the fraction of time the server is free (with zero customers in the system)?

2. A machine has two critically important parts and is subject to three different types of shocks. Shocks of type i occur at times of a Poisson process with rate λ_i. Shocks of types 1 break part 1, those of type 2 break part 2, while those of type 3 break both parts. Let U and V be the failure times of the two parts.

   (a) Find $P(U > s, V > t)$.

   (b) Find the marginal distributions of U and V. Are U and V independent?

3. Consider a system of N components that work independently of each other. Suppose that the lifespan of each component is $\text{Exp}(\lambda)$, and that when a component fails, repair begins immediately and takes a random time with distribution $\text{Exp}(1)$.

   (a) If the number of functioning components is X_t, then $\{X_t\}$ is a CTMC. Find its rate matrix Q.

   (b) Find the stationary distribution.

4. Let $(X_t) =^d \text{Markov}(\lambda, Q)$ be a finite state CTMC. Show that for all $t, h \geq 0$, conditional on $X_t = i$, the random variable X_{t+h} is independent of $(X_s, s \leq t)$, and

$$P(X_{t+h} = j \mid X_t = i) = \delta_{i,j} + q_{i,j}h + o(h),$$

   uniformly in t, as $h \to 0$.

5. (Exercise given in class) Let (X_t) be a CTMC and $\{Y_n\}$ be its embedded DTMC.

   (a) If $i \to j$ in (X_t) then $i \to j$ in $\{Y_n\}$.
   [Hint: First show that $i \to j$ in $\{Y_n\}$ when $P_i(X_n = j \text{ for some } n) > 0$.]

   (b) If state i is recurrent for $\{Y_n\}$ then (X_t) does not explode starting from state i (i.e., $P_i(\zeta = \infty) = 1$).

6. Let $(X_t) =^d \text{Markov}(\lambda, Q)$. Let $T_i = \inf\{t \geq J_1 : X_t = i\}$ be the first passage time to state i. Show that

(a) State i is recurrent if and only if $q_i = 0$ or $P_i(T_i < \infty) = 1$.

(b) State i is transient if and only if $q_i > 0$ and $P_i(T_i < \infty) < 1$.

7. Give an example of an irreducible and recurrent CTMC (X_t) such that

   (a) State i is positive recurrent in (X_t) but not in its embedded chain $\{Y_n\}$.

   (b) State i is positive recurrent in the embedded chain $\{Y_n\}$ but not in the CTMC (X_t).

8. Consider an $M/M/1$ queue with arrival rate λ and departure rate μ. Show that the queue is positive recurrent if and only if $\lambda < \mu$.

9. Let $(X_t) =^d \text{Markov}(\lambda, Q)$. Fix $h > 0$ and consider the DTMC $\{Z_n\}$ defined by $Z_n = X_{nh}$. Show that if Q is irreducible then $\{Z_n\}$ is irreducible and aperiodic.

10. Customers arrive at a single server at rate λ and require an exponential amount of service with rate μ. Customers waiting in line are impatient and if they are not in service they will leave at rate δ independent of their position in the queue. Let X_t denote the number of customers in the system at time t.

    (a) Describe the transition rates of the CTMC (X_t).

    (b) Find the stationary distribution in the very special case in which $\delta = \mu$.

11. Consider a CTMC with state space $\{0, 1, 2\}$ and rate matrix

$$Q = \begin{array}{c} \\ 0 \\ 1 \\ 2 \end{array} \begin{array}{c} \begin{array}{ccc} 0 & 1 & 2 \end{array} \\ \begin{pmatrix} -3 & 1 & 2 \\ 0 & -1 & 1 \\ 2 & 0 & -2 \end{pmatrix} \end{array}$$

Compute $\lim_{t\to\infty} P(X_t = 2 \mid X_0 = 1)$.

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1. (Reading assignment). Example 3.5.4 in Norris.

2. Consider a stationary $M/M/1$ queue with arrival rate λ and service rate μ. Let Y_t denote the number of departures in time $[0, t]$. Using a direct argument (without invoking reversibility), show that $\{Y_t\}$ is a Poisson process of rate λ.

3. ($M/M/\infty$ queue). In the pre-digital days, a phone call required a dedicated line. Suppose that calls arrived at such a facility as a Poisson process with rate λ, and that the length of calls was (independent) exponentially distributed with rate μ.

   (a) Assume that as many lines are available as needed. Let X_t denote the number of occupied lines at time t. Write down the transition rates of the CTMC $\{X_t\}$.

   (b) Show that X_t is is approximately Poisson(λ/μ) for large t.

   (c) Under stationarity, let Y_t denote the number of calls that are completed during time $[0, t]$. What is the distribution of Y_t?

4. Let $\{X_t\}$ be a CTMC with a stationary distribution π that satisfies the detailed balance condition. Let $\{Y_t\}$ be the chain constrained to stay in a subset A of the state space. That is, jumps which take the chain out of A are not allowed, but allowed jumps occur at the original rates. In symbols, $\bar{q}_{x,y} = q_{x,y}$ if $x, y \in A$ and 0 otherwise. Let $C = \sum_{y \in A} \pi(y)$. Show that $\nu(x) = \pi(x)/C$ is a stationary distribution for $\{Y_t\}$.

5. Consider two $M/M/1$ queues in tandem having service rates μ_1 and μ_2. Customers arrive into the first queue according to a Poisson process of rate λ. A customer leaving the first queue joins the second queue with probability p and leaves the system with probability $1 - p$, independent of everything else.

   (a) Under stationarity, what is the arrival process into the second queue?

   (b) Identify a necessary and sufficient condition for stability of the queue lengths, and write down the stationary distribution for the queue length process $\{(X_t^{(1)}, X_t^{(2)})\}$.

   For both parts you should justify your answer (but no computations are needed).

6. Consider the setting of a Jackson network explained in class.

   (a) Show that the traffic equations

$$\beta_i = \lambda_i + \sum_{j=1}^{K} \beta_j p_{i,j}$$

   has a unique non-negative solution.

(b) Show that if $\rho_i = \beta_i/\mu_i < 1$ for all i, then the CTMC $(X_t^{(1)}, \ldots, X_t^{(K)})$ is positive recurrent with stationary distribution

$$\pi(\mathbf{n}) = \prod_{i=1}^{K} (1-\rho_i)\rho_i^{n_i}, \quad \mathbf{n} = (n_1, \ldots, n_K).$$

For the problems below, let $(\Omega, \mathcal{F}, P)$ be a probability space, and let $\mathcal{G} \subset \mathcal{F}$ be a σ-field. All random variables X, X_1, and X_2 below are assumed to be integrable.

7. (Exercise given in class). Suppose (X, Z) has joint density f. Show that

$$Y(\omega) = \begin{cases} \dfrac{\int g(x) f(x, Z(\omega))dx}{\int f(x, Z(\omega))dx} & \text{if the denominator is positive} \\ 0 & \text{otherwise} \end{cases}$$

is a version of $E[g(X) \mid Z]$.

8. If X_1 and X_2 are $\mathcal{F}$-measurable random variables with $X_1 = X_2$ on $B \in \mathcal{G}$, show that $E[X_1 \mid \mathcal{G}] = E[X_2 \mid \mathcal{G}]$ a.s. on B.

9. Show the following properties of conditional expectation.

   (a) Linearity: $E[\alpha X_1 + \beta X_2 \mid \mathcal{G}] = \alpha E[X_1 \mid \mathcal{G}] + \beta E[X_2 \mid \mathcal{G}]$.

   (b) Positivity: If $X_1 \leq X_2$ then $E[X_1 \mid \mathcal{G}] \leq E[X_2 \mid \mathcal{G}]$.

10. Let $\{Y_i\}$ be a sequence of i.i.d. random variables with mean μ and variance σ^2. Let N be a non-negative integer-valued random variable with $E[N^2] < \infty$, independent of $\{Y_i\}$. Let $X = Y_1 + Y_2 \ldots + Y_N$. Show that

    (a) $E[X] = \mu E[N]$.

    (b) $\text{var}(X) = \sigma^2 E[N] + \mu^2 \text{var}(N)$.

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$$\pi(\mathbf{n}) = \prod_{i=1}^{K} (1-\rho_i)\rho_i^{n_i}, \quad \mathbf{n} = (n_1, \ldots, n_K).$$

For the problems below, let $(\Omega, \mathcal{F}, P)$ be a probability space, and let $\mathcal{G} \subset \mathcal{F}$ be a σ-field. All random variables X, X_1, and X_2 below are assumed to be integrable.

7. (Exercise given in class). Suppose (X, Z) has joint density f. Show that

$$Y(\omega) = \begin{cases} \dfrac{\int g(x) f(x, Z(\omega))dx}{\int f(x, Z(\omega))dx} & \text{if the denominator is positive} \\ 0 & \text{otherwise} \end{cases}$$

is a version of $E[g(X) \mid Z]$.

8. If X_1 and X_2 are $\mathcal{F}$-measurable random variables with $X_1 = X_2$ on $B \in \mathcal{G}$, show that $E[X_1 \mid \mathcal{G}] = E[X_2 \mid \mathcal{G}]$ a.s. on B.

9. Show the following properties of conditional expectation.

   (a) Linearity: $E[\alpha X_1 + \beta X_2 \mid \mathcal{G}] = \alpha E[X_1 \mid \mathcal{G}] + \beta E[X_2 \mid \mathcal{G}]$.

   (b) Positivity: If $X_1 \leq X_2$ then $E[X_1 \mid \mathcal{G}] \leq E[X_2 \mid \mathcal{G}]$.

10. Let $\{Y_i\}$ be a sequence of i.i.d. random variables with mean μ and variance σ^2. Let N be a non-negative integer-valued random variable with $E[N^2] < \infty$, independent of $\{Y_i\}$. Let $X = Y_1 + Y_2 \ldots + Y_N$. Show that

    (a) $E[X] = \mu E[N]$.

    (b) $\text{var}(X) = \sigma^2 E[N] + \mu^2 \text{var}(N)$.

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1. Let $(\Omega, \mathcal{F}, P)$ be a probability space. Suppose that $\{B_i\} \subset \mathcal{F}$ forms a disjoint partition of Ω, that is, $\Omega = \bigcup_{i=1}^{\infty} B_i$ and $B_i \cap B_j = \emptyset$ for $i \neq j$. Assume that $P(B_i) > 0$ for all i. Let $\mathcal{G} = \sigma(B_1, B_2, \ldots)$. For an $\mathcal{F}$ measurable, integrable random variable X, show that

$$E[X \mid \mathcal{G}] = \frac{E[X\mathbf{1}_{B_i}]}{P(B_i)} \quad \text{on } B_i.$$

2. Suppose $\{X_n\}$ is a martingale with respect to $\{\mathcal{F}_n\}$. Let $\mathcal{G}_n = \sigma(X_1, \ldots, X_n)$. Then show that $\{X_n\}$ is a martingale with respect to $\{\mathcal{G}_n\}$.

3. Let $\{X_n\}$ be a DTMC with state space $\mathcal{S}$ and transition probability matrix P. Given a function $f : \mathcal{S} \to \mathbb{R}$, consider the operator $\mathcal{L}$ acting on f by

$$(\mathcal{L}f)(x) = \sum_{y \in \mathcal{S}} p_{x,y} \left(f(y) - f(x) \right).$$

That is, $\mathcal{L}f$ is a function on $\mathcal{S}$, and its value at x is given by the above formula. Note that $(\mathcal{L}f)(x) = E[f(X_1) - f(X_0) \mid X_0 = x]$. Define $Y_0 = 0$ and

$$Y_n = f(X_n) - f(X_0) - \sum_{i=0}^{n-1} (\mathcal{L}f)(X_i), \quad n \geq 1.$$

Show that $\{Y_n\}$ is a martingale with respect to $\{\sigma(X_0, \ldots, X_n)\}$.

4. (Exercise given in class). Recall the Galton-Watson branching process $\{X_n\}$: Let $\{Z^n\} = \{(Z_i^n)\}$ be i.i.d. nonnegative integer-valued random variables with mean μ, and define

$$X_{n+1} = \sum_{i=1}^{X_n} Z_i^{n+1}, \quad n \geq 1,$$

and $X_0 = 1$. Show that $\{\mu^{-n} X_n, n \geq 1\}$ is a martingale with respect to $\{\sigma(Z^n)\}$.

5. A superharmonic function $f : \mathbb{R}^d \to \mathbb{R}$ has the following property: for any $x \in \mathbb{R}^d$,

$$f(x) \geq \frac{1}{\text{volume}(B(0,r))} \int_{B(x,r)} f(y) dy,$$

where $B(x,r) = \{y \in \mathbb{R}^d : |x - y| \leq r\}$ denotes the ball of radius r around x. Let $\{Z_n\}$ be i.i.d. $\mathbb{R}^d$-valued random variables uniformly distributed on $B(0,1)$. Let $S_0 = 0$ and define $S_n = S_{n-1} + Z_n$. Define $X_n = f(S_n)$. Show that $\{X_n\}$ is a supermartingale with respect to $\{\sigma(S_1, \ldots, S_n)\}$.

6. Let $\{X_n\}$ be a DTMC with state space $\mathcal{X}$ and transition probabilities $\{p_{x,y}, x, y \in \mathcal{X}\}$. Suppose that $f : \mathcal{X} \to \mathbb{R}$ is a bounded function satisfying

$$\sum_{y \in \mathcal{X}} p_{x,y} f(y) \leq \lambda f(x)$$

for some $\lambda > 0$ and all $x \in \mathcal{X}$. Show that $\{\lambda^{-n} f(X_n)\}$ is a supermartingale with respect to $\{\sigma(X_0, \ldots, X_n)\}$.

7. (Polya's urn). A bag contains red balls and blue balls, with initially r red balls and b blue balls (both r and b are nonzero). A ball is drawn from the bag, its color is noted, and it is returned to the bag together with a new ball of the same color. Let R_n denote the number of red balls after n such operations.

   (a) Let

   $$Y_n = \frac{R_n}{n + r + b}, \quad n \geq 0.$$

   Show that $\{Y_n\}$ is a martingale with respect to $\{\sigma(R_1, \ldots, R_n)\}$.

   (b) Let T be the number of balls drawn until the first blue ball appears, and suppose that $r = b = 1$. Show that $E[(T + 2)^{-1}] = \frac{1}{4}$.

8. Let $\{Z_i\}$ be i.i.d. with $P(Z_1 = 1) = P(Z_1 = -1) = 1/2$, and consider the symmetric random walk defined by $X_0 = 0$ and $X_n = \sum_{i=1}^{n} Z_i$. Fix positive integers a and b, and consider the hitting time $T = \min\{n \geq 0 : X_n \in \{-a, b\}\}$. By identifying suitable martingales,

   (a) Show that $P(X_T = -a) = \frac{b}{a+b}$;

   (b) Show that $E[T] = ab$.

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1. Consider a DTMC $\{X_n\}$ and suppose state N is such that $p_{N,N} = 0$. Let $g(i)$ denote the probability that this chain eventually enters state N given that it starts in state i. Show that $\{g(X_n)\}$ is is a martingale with respect to $\{\sigma(X_1, \ldots, X_n)\}$.

2. Given an example of i.i.d. integrable random variables $\{X_1, \ldots, X_n\}$ and a random time N such that the conclusion of Wald's lemma fails: $E\left[\sum_{n=1}^{N} X_n\right] \neq E[N]E[X_1]$.

3. (Exercise given in class) Let X be a random variable with $-a \leq X \leq b$ and $E[X] = 0$. If f is a convex function, show that

$$E[f(X)] \leq \frac{1}{a+b}(bf(-a) + af(b))$$

4. (Exercise given in class) Let $\{X_n\}$ be the Galton-Watson branching process discussed in class. If $\mu = 1$ and $P(Z_1^1 = 1) < 1$, then show that the population dies out almost surely.

5. Let $\{X_1, \ldots, X_n\}$ be independent random vectors uniformly distributed on the circle of radius 1 centered at the origin. Let $T = T(X_1, \ldots, X_n)$ denote the length of the shortest path connecting these n points. Show that

$$P(|T - E[T]| \geq a) \leq 2\exp\{-a^2/32\}.$$

6. For binary vectors x and y of length n, let $d_H(x, y) = \sum_{i=1}^{n} |x_i - y_i|$ denote the Hamming distance between x and y, that is, number of positions in which x and y differ. Let A be a finite set of vectors, and let $X = (X_1, \ldots, X_n)$, where X_i are i.i.d. Bernoulli(1/2) random variables. Let $D = \min_{y \in A} d_H(X, y)$, and let $\mu = E[D]$. For $a > \mu$, find an upper bound for $P(D \geq a)$.

7. Consider a sequence of random variables $\{X_n\}$ defined as follows. Let $X_1 = 1$. Given $X_1, \ldots, X_{n-1}$, define X_n to be a Poisson random variable with parameter X_{n-1}. What can you say about this sequence when n is large?

Industrial Engineering and Operations Research
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IE 622: Probability and Stochastic Processes II
Quiz 1

Name:	Date: Jan 8, 2026
ID:	Marks obtained:
Department:	Max. marks: 10

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space.

Indicate whether the following statements are TRUE/FALSE (0.5 marks) and justify your answer (1.5 marks). Justification is not evaluated if the TRUE/FALSE answer is wrong.

1. If $(A_n)_{n\geq 1}$ is a decreasing sequence of sets in $\mathcal{F}$ i.e.,

$$A_1 \supseteq A_2 \supseteq \cdots \supseteq A_n \supseteq \cdots,$$

   then $\mathbb{P}(A_n) \downarrow \mathbb{P}\big(\cap_{n=1}^{\infty} A_n\big)$.

2. If X, Y are integer-valued and have probability mass functions (pmf) f and g, then $X+Y$ has pmf $h(n) := \sum_{k\in\mathbb{Z}} f(k)g(n-k)$.

3. Two random variables X and Y are independent if $\mathbb{E}[XY] = \mathbb{E}[X]\,\mathbb{E}[Y]$.

4. Let $X \sim \text{Exp}(1)$. The conditional probability $P(X > t+s \mid X > s)$ is strictly less than the unconditional probability $P(X > t)$ because the total time $t+s$ is longer than t.

5. Let $X \sim \text{Exp}(1)$ and $Y := 1 - e^{-X}$. Then $Y \sim \text{Unif}([0,1])$.

Industrial Engineering and Operations Research
IIT Bombay
IE 622: Probability and Stochastic Processes II
Quiz 2

Name: Date: Jan 15, 2026
ID: Marks obtained:
Department: Max. marks: 10

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1. Let $(X_n)_{n\geq 0}$ be an i.i.d. binary sequence with $X_n \sim \text{Bernoulli}(p)$. A technician (same person as in the problem set) alters an outdated hardware to make a new device called the `RunLengthLimiterPro`, that modifies the input $(X_n)_{n\geq 0}$ to produce an output sequence $(Y_n)_{n\geq 0}$ as follows:

   (a) Initialization: $Y_0 = X_0, \qquad Y_1 = X_1, \qquad Y_2 = X_2.$

   (b) For all $n \geq 3$:

   - If $Y_{n-1} = Y_{n-2} = Y_{n-3}$, then $Y_n = 1 - Y_{n-1}$.
   - If $Y_{n-1} = Y_{n-2} \neq Y_{n-3}$, then

   $$Y_n = \begin{cases} 1 - X_n, & \text{with probability } p, \\ X_n, & \text{with probability } 1-p. \end{cases}$$

   - Otherwise, $Y_n = X_n$.

   Model the output process (Y_n) as a Markov chain by:

   (a) specifying a suitable state space;

   (b) giving the transition probabilities.

   (c) drawing the transition probability diagram.

Industrial Engineering and Operations Research
IIT Bombay
IE 622: Probability and Stochastic Processes II
Quiz 3

Name: Date: Jan 22, 2026
ID: Marks obtained:
Department: Max. marks: 10

Let $(X_n)_{n\geq 0}$ be a Markov chain on $I = \{1, 2, 3, 4, 5\}$ with transition probability matrix

$$P = \begin{pmatrix} 0.1 & 0.8 & 0 & 0 & 0.1 \\ 0.3 & 0.7 & 0 & 0 & 0 \\ 0 & 0 & 0.9 & 0.1 & 0 \\ 0 & 0 & 0.4 & 0.6 & 0 \\ 0 & 0.1 & 0.8 & 0.1 & 0 \end{pmatrix}$$

For $A \subset I$, let $H^A = \inf\{n \geq 0 : X_n \in A\}$ be the hitting time of A and let $h_i^A := \mathbb{P}(H^A < \infty | X_0 = i)$ be the hitting probability to A starting from i.

1. (1 mark) Find the communicating classes and identify a closed communicating class C. (There is at least one since the state space is finite)
2. (1.5 marks) Find h_i^C for all $i \in I$.
3. (1.5 marks) Let $D = I \setminus C$. Find h_i^D for all $i \in I$.
4. (6 marks) For all $i, j \in D$ such that $i \neq j$, find $h_i^j \equiv h_i^{\{j\}}$.

Industrial Engineering and Operations Research
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IE 622: Probability and Stochastic Processes II
Quiz 4

Name:	Date: Jan 29, 2026
ID:	Marks obtained:
Department:	Max. marks: 10

Let $(X_n)_{n\geq 0}$ be a discrete-time Markov chain with transition matrix $P = (p_{ij})$ and state space I. Define the first return time to state j by

$$T_j := \inf\{n \geq 1 : X_n = j\},$$

and let $f_{jj} := \mathbb{P}_j(T_j < \infty)$. Let $N_{j\to k}$ be the number of transitions from state j to state k. Show that the expected value of $N_{j\to k}$ given that the Markov chain starts at $X_0 = j$ is given by

$$\mathbb{E}_j\left[N_{j\to k}\right] = \frac{p_{jk}}{1 - f_{jj}}.$$

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Quiz 5

Name: Date: Feb 5, 2026
ID: Marks obtained:
Department: Max. marks: 10

Consider the DTMC with state space $\{0, 1, \dots, \}$ and transition probabilities

$$p_{i,j} = \begin{cases} 1 & \text{if } i \geq 1 \text{ and } j = i - 1, \\ q_k & \text{if } i = 0 \text{ and } j = k \geq 1, \\ q_0 & \text{if } i = 0 \text{ and } j = 0, \\ 0 & \text{otherwise,} \end{cases}$$

where $\{q_i, i = 0, 1, 2, \dots\}$ are such that $0 < q_i < 1$ and $\sum_{i=0}^{\infty} q_i = 1$.

1. Show that this DTMC is recurrent.
2. Find the mean recurrence time of state 0 and identify when the DTMC is positive recurrent.
3. If $\pi = (\pi_0, \pi_1, \dots)$ is the stationary distribution when it is positive recurrent, then find π_0.
4. Is the DTMC aperiodic? Justify your answer.
5. If $q_{2m} = 0$ for $m = 0, 1, 2, \dots$, then find the period of state 0 and state 10.

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Quiz 6

Name:	Date: Feb 12, 2026
ID:	Marks obtained:
Department:	Max. marks: 10

Theorem 0.1 (Foster-Lyapunov Criteria) *Let $(X_n)_{n\geq 0}$ be an irreducible Markov chain on a countable state space $\mathcal{S} = \{0, 1, 2, 3, \cdots\}$. Let $V : \mathcal{S} \to [0, \infty)$ be a Lyapunov function and $A \subset \mathcal{S}$ be a finite set. Define the drift operator $\Delta V(x) = E[V(X_{n+1}) - V(X_n) \mid X_n = x]$.*

1. ***Positive Recurrence:*** *If $\Delta V(x) \leq -\epsilon$ for all $x \notin A$ (for some $\epsilon > 0$) and $E[V(X_{n+1}) \mid X_n = x] < \infty$ for all $x \in A$, then (X_n) is positive recurrent.*

Let $p, q > 0$. Consider a DTMC with the entries of the transition probability matrix $P = (p_{ij})_{i,j\in\mathcal{S}}$ given by:

$$p_{i,i+2} = 1 - p_{i,i-1} = p \qquad \text{for all } i \geq 1, \qquad \text{and} \qquad p_{0,2} = 1 - p_{0,0} = q.$$

- Obtain a sufficient condition for positive recurrence of this DTMC using a linear Lyapunov function $V(x) = x$.
- Let $V'(x) = x^2$. How does the condition for positive recurrence obtained using $V'(x)$ compare with the one in the previous part?

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Quiz 7

Name: Date: Feb 19, 2026
ID: Marks obtained:
Department: Max. marks: 10

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space. Let $(A_n)_{n\geq 1}$ be a sequence of events in $\mathcal{F}$. Recall the following:

- $\liminf A_n = \{\omega \in \Omega : \omega \text{ belongs to all but finitely many } A_n\}$.
- **Fatou's lemma**: For non-negative random variables $(X_n)_{n\geq 1}$, $\mathbb{E}[\liminf X_n] \leq \liminf \mathbb{E}[X_n]$.

Answer the following. Justify all the steps.

1. Show that $\liminf A_n \in \mathcal{F}$.
2. Show that $\mathbb{1}_{\liminf A_n}(\omega) = \liminf \mathbb{1}_{A_n}(\omega)$, for any $\omega \in \Omega$.
3. Show that $\mathbb{P}(\liminf A_n) \leq \liminf \mathbb{P}(A_n)$.

IE 622 Quiz 8 Solution

Problem 1

1. The random variables X_s and X_t are independent.

Answer: $\boxed{\text{False}}$
Justification: In a Poisson process, X_t represents the total number of events from time 0 up to time t. Because the interval $(0, t]$ fully contains $(0, s]$, we can write X_t as the sum of events up to time s and events between s and t:

$$X_t = X_s + (X_t - X_s)$$

Since X_t explicitly includes the value of X_s, their values are dependent. For instance, knowing $X_s = 5$ guarantees that $X_t \geq 5$, meaning $\boxed{P(X_t < 5 \mid X_s = 5) = 0}$, whereas the unconditional probability is $\boxed{P(X_t < 5) > 0}$.

2. The random variables $X_t - X_s$ and X_{t-s} are the same.

Answer: $\boxed{\text{False}}$
Justification: While they are **identically distributed**, they are not the **exact same** random variable. $X_t - X_s$ counts the specific arrivals in the time interval $\boxed{(s, t]}$, whereas X_{t-s} (which is $X_{t-s} - X_0$) counts the arrivals in the time interval $\boxed{(0, t-s]}$. A specific realization (sample path) of the process could easily have different counts in these two distinct time intervals. However, due to the stationary increments property of the Poisson process, both variables follow the exact same distribution: Poisson$(t - s)$.

Problem 2

Let $(X_t)_{t \geq 0}$ be a counting process ($X_0 = 0$) with independent and stationary increments, with $X_t \sim \text{Poisson}(\lambda t)$ for all $t > 0$. Given $0 \leq t_1 < t_2 < \cdots < t_n$ and $0 \leq x_1 \leq x_2 \cdots \leq x_n$.

We need to compute $P(X_{t_1} = x_1, X_{t_2} = x_2, \ldots, X_{t_n} = x_n)$. Let $t_0 = 0$ and $x_0 = 0$. We transform the event into a sequence of non-overlapping increments:

$$P(X_{t_1} = x_1, \ldots, X_{t_n} = x_n) = \boxed{P\left(\bigcap_{i=1}^{n} \{X_{t_i} - X_{t_{i-1}} = x_i - x_{i-1}\}\right)}$$

Because the process has $\boxed{\textbf{independent increments}}$, the joint probability of these non-overlapping intervals factors into a product:

$$= \boxed{\prod_{i=1}^{n} P(X_{t_i} - X_{t_{i-1}} = x_i - x_{i-1})}$$

By the $\boxed{\textbf{stationary increments}}$ property, each increment $X_{t_i} - X_{t_{i-1}}$ follows a Poisson$(\lambda(t_i - t_{i-1}))$ distribution. The PMF for each increment is:

$$P(X_{t_i} - X_{t_{i-1}} = x_i - x_{i-1}) = \frac{e^{-\lambda(t_i - t_{i-1})}(\lambda(t_i - t_{i-1}))^{x_i - x_{i-1}}}{(x_i - x_{i-1})!}$$

Substituting this into our product yields:

$$\begin{aligned} P(X_{t_1} = x_1, \ldots, X_{t_n} = x_n) &= \prod_{i=1}^{n} \frac{e^{-\lambda(t_i - t_{i-1})}(\lambda(t_i - t_{i-1}))^{x_i - x_{i-1}}}{(x_i - x_{i-1})!} \\ &= e^{-\lambda \sum_{i=1}^{n}(t_i - t_{i-1})} \prod_{i=1}^{n} \frac{(\lambda(t_i - t_{i-1}))^{x_i - x_{i-1}}}{(x_i - x_{i-1})!} \\ &= \boxed{e^{-\lambda t_n} \prod_{i=1}^{n} \frac{(\lambda(t_i - t_{i-1}))^{x_i - x_{i-1}}}{(x_i - x_{i-1})!}} \end{aligned}$$

Part 2: Finite-Dimensional Distributions Completely Determined

The finite-dimensional distributions of a process define the joint probability at any finite sequence of times $t_1, \ldots, t_n$. To say they are completely determined means we can state the exact probability law for any arbitrary set of inputs. As derived above, the joint probability depends exclusively on the provided times t_i, the states x_i, and the fundamental rate λ. Since we have an exact, $\boxed{\text{closed-form formula to compute this joint probability without any}}$ $\boxed{\text{undefined parameters}}$, the finite-dimensional distributions are completely characterized.

Marking Scheme

Problem 1 (4 Marks)

- [**1 Mark**] Correctly identifying Statement 1 as False.
- [**1 Mark**] Valid justification.
- [**1 Mark**] Correctly identifying Statement 2 as False.
- [**1 Mark**] Valid justification.

Problem 2 (6 Marks)

[**1 Mark**] for each box.

IE 622, Quiz 9. Mar 12, 2026. Name: Roll No:

Problem 1: Let (X_t) be a Poisson process of rate λ. For any $t > 0$, show that

$$\frac{X_{nt}}{n} \to \lambda t \quad \text{as } n \to \infty, \quad \text{almost surely.}$$

(6 marks)

Problem 2: State true or false, and justify your answer.

1. The even arrivals (i.e., the second arrival, fourth arrival, etc.) in a Poisson process of rate λ constitute a Poisson process of rate $\lambda/2$.
2. Fix $0 < s < t$. The location of arrivals in a Poisson process of unit rate in the interval $[s, t]$ is uniformly distributed.

(2 marks)

Problem 3: Fill in the blanks.

1. For a CTMC with rate matrix Q, the holding time in state i has exponential distribution with parameter _____.
2. For an $M/M/1$ queue with arrival rate λ and departure rate μ, the one-step transition probability of the embedded chain going from 1 to 2 is _____.

(2 marks)

Your answer:

IE 622 Quiz 10 Solution

Problem 1

Because the given CTMC has a finite state space and is irreducible (it is possible to transition between all states, e.g., $0 \to 1 \to 2 \to 0$), it has a unique stationary distribution $\pi = (\pi_0, \pi_1, \pi_2)$. For such a chain, the limiting probabilities exist and are independent of the initial state.

Therefore, $\lim_{t\to\infty} P(X_t = 2 \mid X_0 = 1) = \pi_2$.

To find the stationary distribution, we solve the global balance equations $\pi Q = \mathbf{0}$ alongside the normalization condition $\sum \pi_i = 1$:

$$(\pi_0, \pi_1, \pi_2) \begin{pmatrix} -3 & 2 & 1 \\ 0 & -1 & 1 \\ 1 & 0 & -1 \end{pmatrix} = (0, 0, 0)$$

This yields the following system of linear equations:

$$\begin{aligned} -3\pi_0 + 0\pi_1 + 1\pi_2 = 0 &\implies \pi_2 = 3\pi_0 \\ 2\pi_0 - 1\pi_1 + 0\pi_2 = 0 &\implies \pi_1 = 2\pi_0 \\ 1\pi_0 + 1\pi_1 - 1\pi_2 = 0 &\quad \text{(redundant)} \end{aligned}$$

Applying the normalization condition ($\pi_0 + \pi_1 + \pi_2 = 1$):

$$\begin{aligned} \pi_0 + 2\pi_0 + 3\pi_0 &= 1 \\ 6\pi_0 &= 1 \\ \pi_0 &= \frac{1}{6} \end{aligned}$$

Substituting π_0 back to find π_2:

$$\pi_2 = 3\left(\frac{1}{6}\right) = \frac{1}{2}$$

Hence: $\lim_{t\to\infty} P(X_t = 2 \mid X_0 = 1) = \pi_2 = \frac{1}{2}$.

Problem 2.1

Answer: True
Justification 1:

1. For a finite, irreducible CTMC, there exists a unique, strictly positive stationary distribution π ($\pi_i > 0$ for all i).

2. In the CTMC: The expected return time is $\mu_i = \frac{1}{\pi_i v_i}$. Since the chain is finite and irreducible, exit rates are bounded ($0 < v_i < \infty$). Therefore, $\mu_i < \infty$ (Positive Recurrent).

3. In the Embedded DTMC: Its stationary distribution is $y_i = \frac{\pi_i v_i}{\sum_j \pi_j v_j}$. Since both π_i and v_i are strictly positive, $y_i > 0$. The expected return steps are $m_i = \frac{1}{y_i} < \infty$ (Positive Recurrent). Hence, state i is mathematically guaranteed to be positive recurrent in both the continuous and embedded discrete chains.

Justification 2:

1. The total continuous time it takes to return to state i (μ_i) is simply the expected number of discrete jumps (m_i) multiplied by the average time spent waiting in each state.

2. Because the chain has a finite number of states and is irreducible, the expected waiting time in any state is strictly bounded between some positive minimum (t_{min}) and a finite maximum (t_{max}).

3. This means the total return time (μ_i) is directly bounded by the number of jumps (m_i):
$$m_i \cdot t_{min} \leq \mu_i \leq m_i \cdot t_{max}$$
Because t_{min} and t_{max} are finite constants, μ_i is finite if and only if m_i is finite. Therefore, the CTMC is positive recurrent ($\mu_i < \infty$) if and only if the embedded DTMC is positive recurrent ($m_i < \infty$).

Problem 2.2

Answer: False
Justification:

1. For $\{Z_n\}$ to be a Discrete-Time Markov Chain
$$P(Z_{n+1} = j \mid Z_n = i, Z_{n-1} = k, \ldots) = P(Z_{n+1} = j \mid Z_n = i)$$
must hold for all i, j and any history of states.

2. The essential issue is that the sequence of stopping times $\{T_n\}$ can be defined in a way that depends on the history of the process i.e. T_n can be a function of the past states (dynamic stopping times).

3. **Counterexample:** Consider a CTMC with states $\{0, 1\}$ and transition rates $q_{01} = q_{10} = 1$. Define $T_0 = 0$, $T_1 = 1$, $T_2 = 2$, and

$$T_3 = \begin{cases} 3 & \text{if } Z_1 = 0 \\ 4 & \text{if } Z_1 = 1 \end{cases}$$

Here, T_3 depends on the initial state Z_0. In this case,

$$P(Z_3 = j \mid Z_2 = 1, Z_1 = 1) \neq P(Z_3 = j \mid Z_2 = 1, Z_1 = 0)$$

Thus, the Markov property is violated, and $\{Z_n\}$ is not a DTMC.

Partial Marking: If a student answers **True**, they will get **1 Mark** if they clearly mentions the **Strong Markov Property** alongwith appropriate explanation.

Marking Scheme

Problem 1 (5 Marks)

- [**1 Mark**] Correctly identifying that the limit equals the stationary probability π_2 due to the **finite**, **irreducible** nature of the CTMC. No marks if the limit is directly equated to π_2 without justification.
- [**1 Marks**] Setting up the correct system of global balance equations $\pi Q = 0$.
- [**1 Mark**] Solving the system to express π_1 and π_2 in terms of π_0.
- [**1 Mark**] Using the normalization condition $\sum \pi_i = 1$ to solve the system.
- [**1 Mark**] Arriving at the correct final answer of 1/2.

Problem 2 (5 Marks)

- [**1 Mark**] Correctly identifying Statement 1 as True.
- [**1.5 Marks**] For each point in the justification (whether justification 1 or 2) awarding 0.5 marks. Mixing points from both justifications will not earn marks. Qualitative arguments will suffice (exact formulae may not be written).
- [**1 Mark**] Correctly identifying Statement 2 as False **provided** correct justification is given (even partially explained is ok but incorrect justification won't suffice).
- [**1.5 Marks**] For each point in the justification awarding 0.5 marks.

IE 622, Quiz 11. Apr 2, 2026. Name: Roll No:

Problem 1: Let $(\Omega, \mathcal{F}, P)$ be a probability space, and let $\mathcal{G} \subset \mathcal{F}$ be a σ-field. Let X_1 and X_2 be integrable random variables defined on this probability space. If $X_1 \leq X_2$, show that $E[X_1 \mid \mathcal{G}] \leq E[X_2 \mid \mathcal{G}]$. (6 marks)

Problem 2:

1. Show that any irreducible CTMC whose state space consists of two elements is reversible with respect to its stationary distribution. (2 marks)
2. Give an example of an irreducible CTMC that is not reversible. (2 marks)

IE 622, Quiz 12. Apr 9, 2026. Name: Roll No:

Problem 1: Let $\{Z_i\}$ be i.i.d. with $P(Z_1 = 1) = P(Z_1 = -1) = 1/2$, and consider the symmetric random walk defined by $X_0 = 0$ and $X_n = \sum_{i=1}^{n} Z_i$. Fix positive integers a and b, and consider the hitting time $T = \min\{n \geq 0 : X_n \in \{-a, b\}\}$. Show that $P(X_T = -a) = b/(a+b)$. (4 marks)

Problem 2: Let $\{X_n\}$ be a submartingale with respect to $\{\mathcal{F}_n\}$. State whether each statement below is true or false, and justify.

1. If $\{Y_n\}$ is a submartingale with respect to $\{\mathcal{F}_n\}$ then $\{X_n + Y_n\}$ is a submartingale with respect to $\{\mathcal{F}_n\}$. (3 marks)
2. If $\{H_n\}$ is predictable then $\{(H \cdot X)_n\}$ is a submartingale with respect to $\{\mathcal{F}_n\}$. (3 marks)

IE 622, Quiz 13. Apr 16, 2026. Name: Roll No:

Problem 1: Consider a sequence of random variables $\{X_n\}$ defined as follows. Let $X_1 = 1$. Given $X_1, \ldots, X_{n-1}$, define X_n to be a Poisson random variable with parameter X_{n-1}. What can you say about this sequence when n is large? (4 marks)

Problem 2: State whether each statement below is true or false, and justify.

1. If $\{X_n\}$ is a submartingale and φ is convex then $\{\varphi(X_n)\}$ is a submartingale. (3 marks)
2. If $\{X_n\}$ is a martingale with $\sup_{n\geq 1} E[|X_n|] < \infty$ then X_n converges in L^1. (3 marks)